

Dataset & Data Specification

ServiceGuard AI

1. Dataset design

The dataset should simulate realistic SaaS/service telemetry over time. It must contain normal periods, gradual degradation, abrupt incidents, recovery periods and enough labeled history to train a supervised risk model.

2. Core telemetry fields

Field	Type	Meaning
service_id	UUID/string	Logical service identifier
customer_id	UUID/string	Customer owning/using service
timestamp	timestamp	Observation time
request_count	integer	Requests in interval
response_time_ms	float	Observed latency
error_rate	float	Fraction/percentage of failed requests
cpu_usage_pct	float	CPU utilization
memory_usage_pct	float	Memory utilization
db_connections_pct	float	Connection pool utilization
network_latency_ms	float	Network latency
availability_pct	float	Availability in interval
sla_target_ms	float	Permitted latency target
incident_next_30m	boolean	Training label
incident_type	categorical	Optional incident category

3. Master data

- customers
- applications
- services
- service_instances
- sla_policies
- dependencies

4. Synthetic-data scenarios

- Normal operation.
- CPU saturation.
- Database connection exhaustion.
- Latency degradation.
- Error-rate spike.
- Network degradation.
- Deployment-related regression.

- Recovery after remediation.

5. Feature engineering

- Rolling mean/max/min.
- Latency slope.
- Error-rate delta.
- Threshold deviation.
- Recent incident count.
- Time since last incident.
- Dependency health indicators.
- Composite resource pressure.

6. Data quality rules

- No impossible percentages.
- Timestamp ordering checks.
- Required identifiers cannot be null.
- Duplicate telemetry policy must be explicit.
- Training labels must be generated from future windows to avoid leakage.